

EDA for Credit Risk Assessment and Loan Defaults

Diving deep into loan application trends and data visualization for better understanding and engagement.

Prepared by [KARTHIKEIN MR]
C72 BATCH

REGISTERED EMAIL ID
kushikarthikeyenmr@gmail.com

Introduction

- This analysis focuses on applications of clients and previous applications of clients of default and non-default clients.
- Understanding Loan Application Trends and Insights
- Datasets are used in my assignment:
- `application_data.csv` is the current loan application.
- `previous_application.csv` comprises all historical data.
- Historical dataset `previous_application.csv` is used to compare current applications with past trends.

Used Libraries my assignment with Function of using These libraries :

➤ **pandas as pd:**

Loads necessary data to import datasets, clean, or transform datasets in this case - CSV files.

➤ **numpy as np:**

The following does operations on large data arrays efficiently; it supports mathematical operations when doing an analysis.

➤ **matplotlib.pyplot as plt:**

Used for plotting graphs (like histograms, bar charts, scatter plots) for data visualization.

➤ **seaborn as sns:**

It allows for easy plotting of complex visualizations such as heatmaps, pair plots, etc., and comes with built-in support for beautiful aesthetics.

➤ **Warnings:**

It's used in this code to suppress warnings that may be printed during execution, for instance, due to missing values or deprecated functions.

Dataset Overview

- The analysis prints the first 5 rows of the data sets.
- Data sets initial shape:
- Application Data[default dataset]
- Previous Application Data[default dataset]
- **Key columns in application_data:**
- SK_ID_CURR: Unique identifier for each applicant.
- TARGET: Binary target variable indicating whether the loan was defaulted (1) or not (0).
- AMT_INCOME_TOTAL: Total annual income of the applicant.
- AMT_CREDIT: The credit amount requested by the applicant.
- CODE_GENDER: Gender of the applicant.
- NAME_CONTRACT_TYPE: Type of loan contract (e.g., cash loans, revolving loans).

- **Key columns in previous_application_data:**
- SK_ID_PREV: Unique identifier for each previous loan application.
- AMT_APPLICATION: Amount requested in the previous loan application.
- AMT_CREDIT: The credit amount granted in the previous loan.
- WEEKDAY_APPR_PROCESS_START: The weekday when the application process started.
- ORGANIZATION_TYPE: The type of organization to which the applicant is affiliated.

Data Cleaning

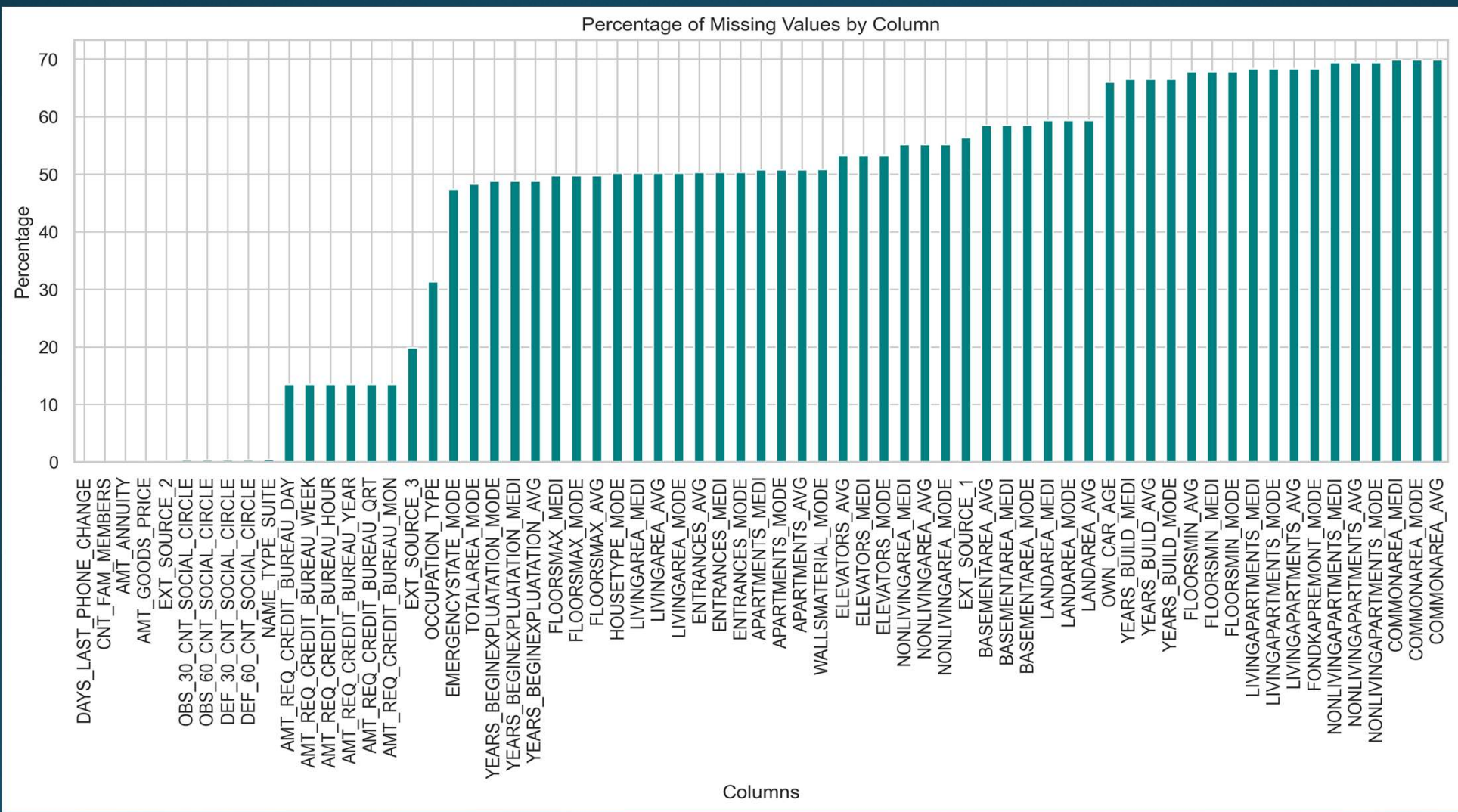
steps in data cleaning process:

- Removed columns containing 'flag' (case-insensitive).
- Handled missing values by:
- Dropping columns with >20% missing values.
- Filling others with median/mode.

Final dataset shapes:

- Application Data: [new shape]
- Previous Application Data: [new shape]

Analyzing and Visualizing Missing Data in Application Data



Income Distribution Analysis

Univariate Analysis of AMT_INCOME_TOTAL



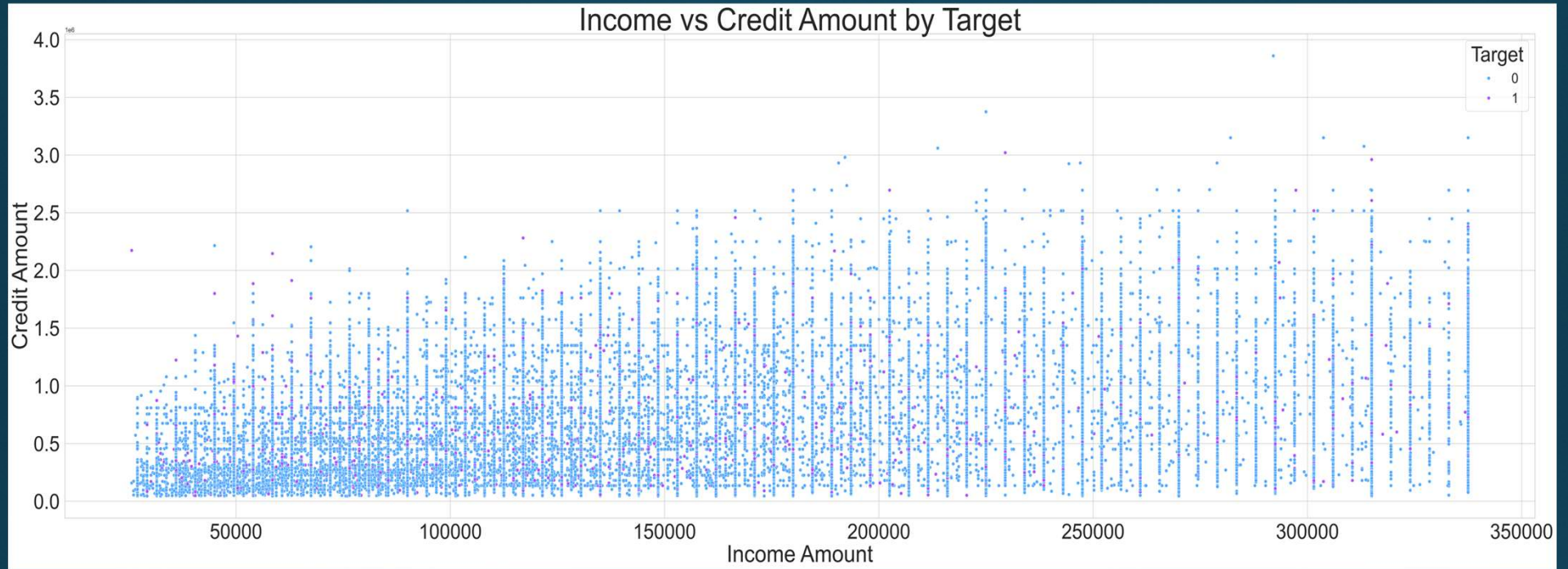
Income_distribution:

This represents the distribution of the entire income,

AMT_INCOME_TOTAL, of all borrowers.

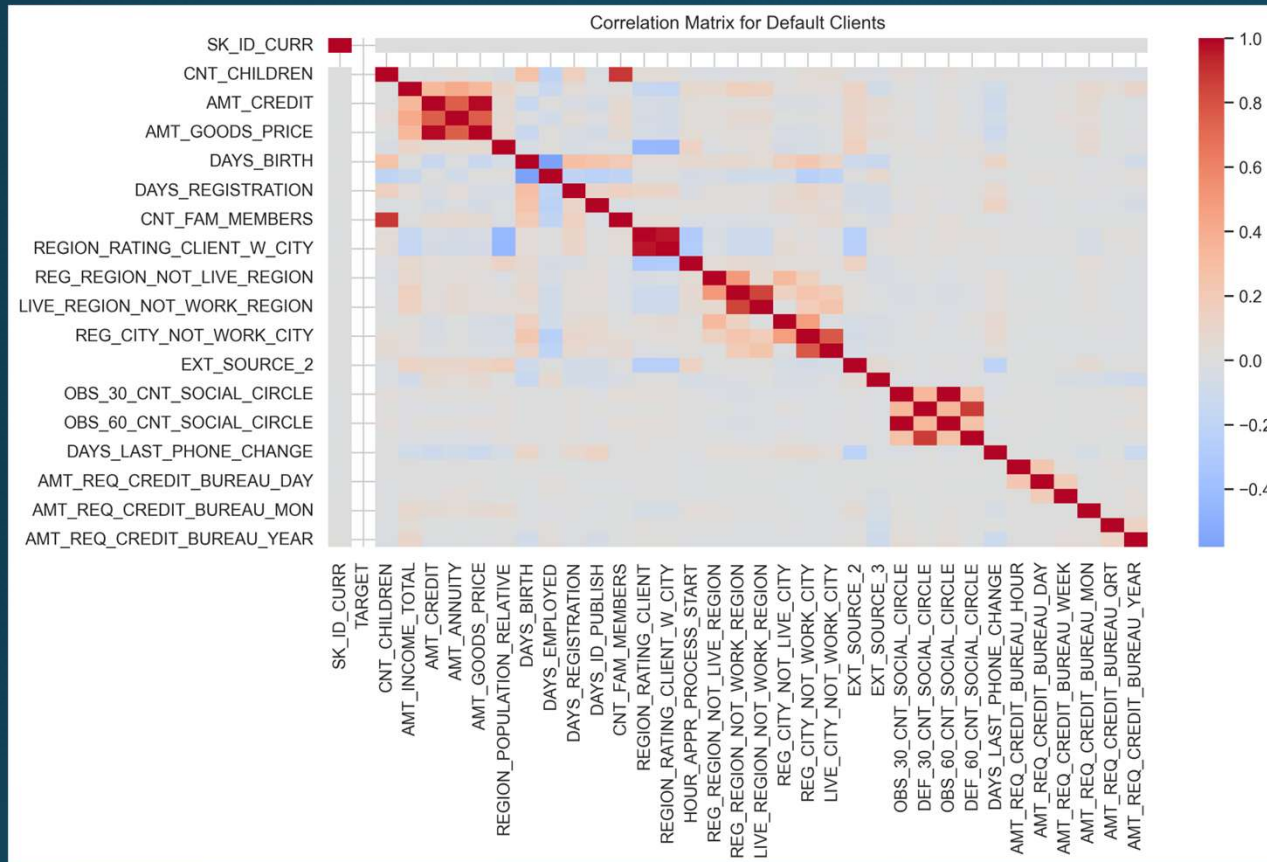
The histogram depicts the relative frequency of each income class

income_vs_credit



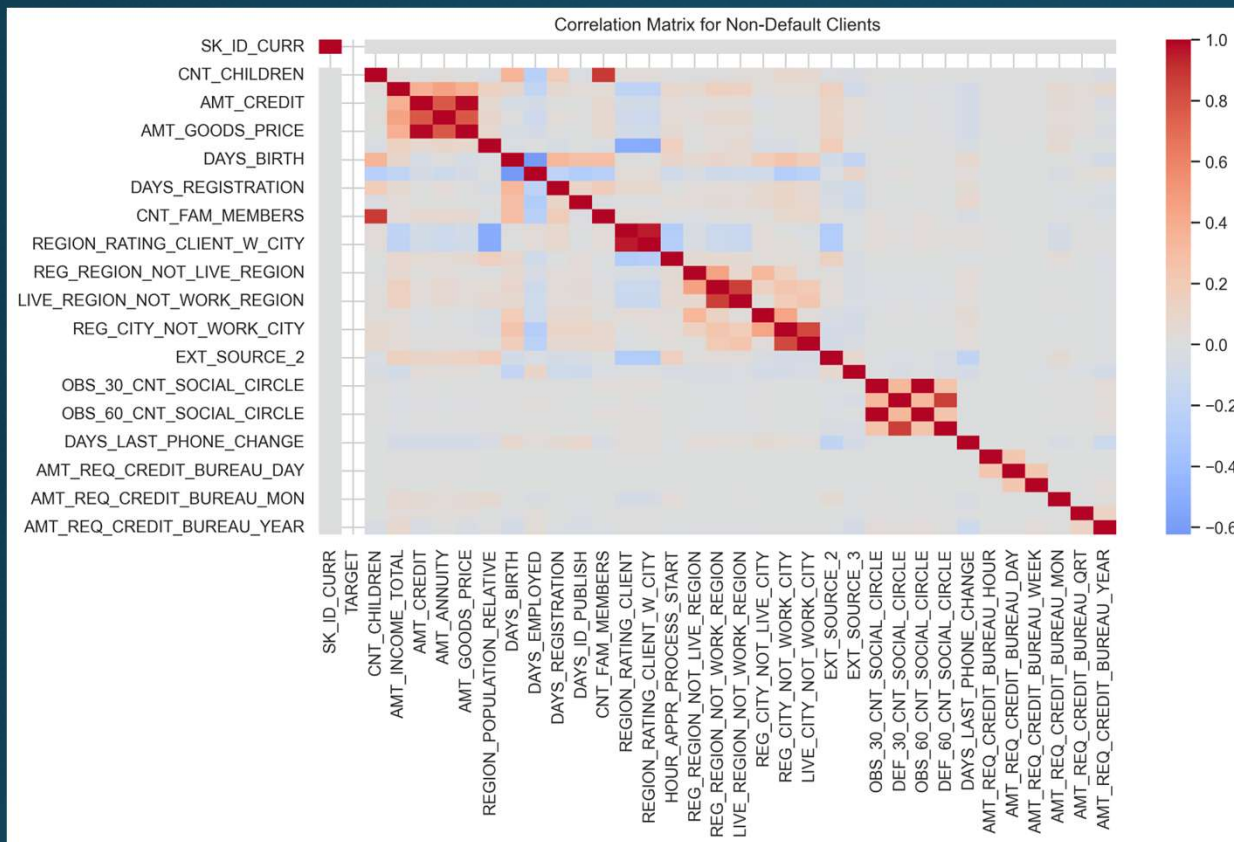
- It is a scatterplot showing a relationship between applicants' income and amounts of credit granted.
- Scatterplot is colored for the target variable:
 - 0: Non-default customers didn't default on loans
 - 1: Default customers defaulted on loans.

correlation matrix for default clients



- Shows relationships between numerical variables for clients who defaulted (TARGET= 1).
- Strong positive correlation observed between AMT_CREDIT and AMT_GOODS_PRICE.
- Useful for identifying risk factors specific to default clients.

correlation matrix for non default clients



- Highlights variable relationships for clients who did not default (TARGET = 0).
- Correlations like AMT_ANNUIITY and AMT_CREDIT remain consistently strong.
- Helps in comparing behavior patterns with default clients.

target distribution



- **Customer Segmentation:**

The graph represents a count of customers split along their credit payment status (TARGET variable).

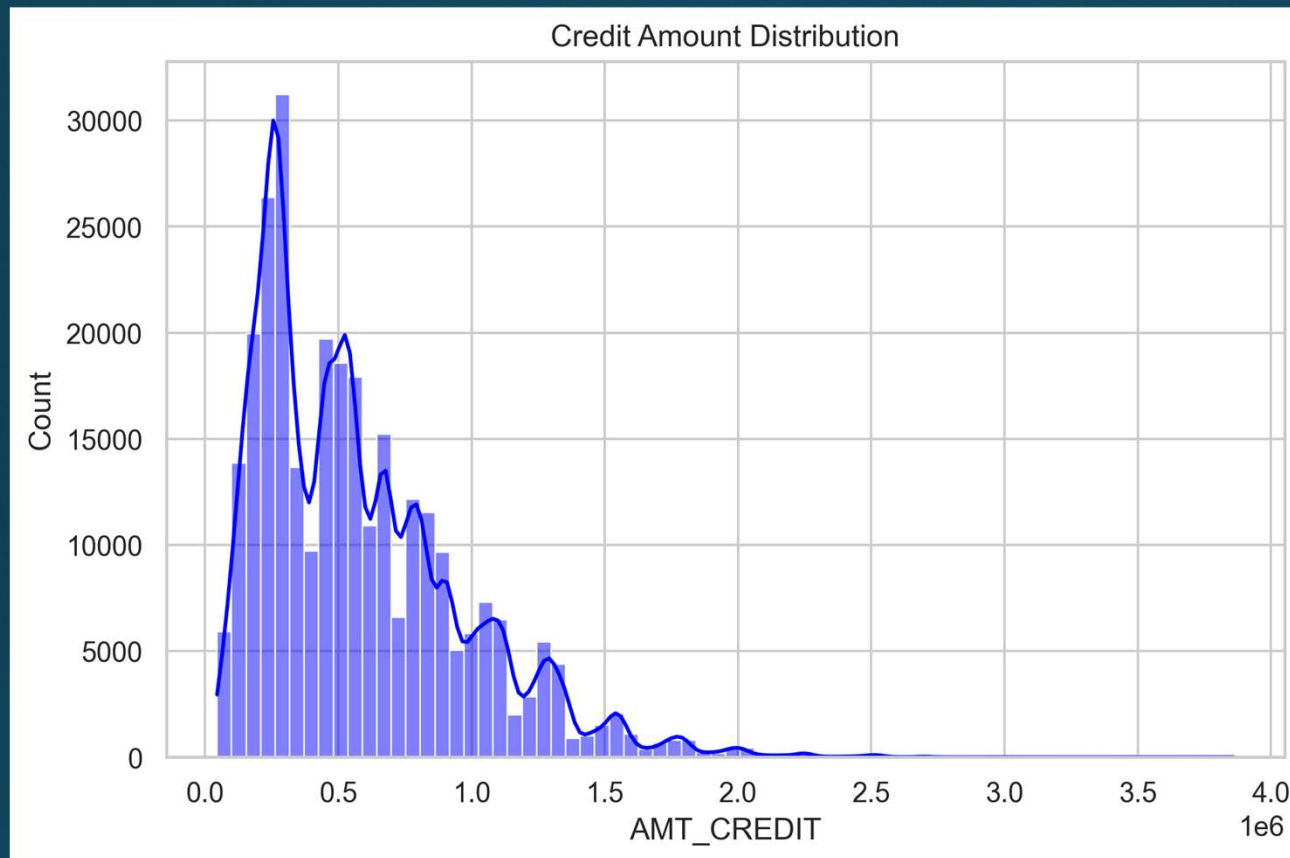
- **Observation of Class Imbalance:**

Most customers belong to the non-defaulted category (TARGET = 0), which reflects a significant class imbalance in the dataset.

- **Consequence:**

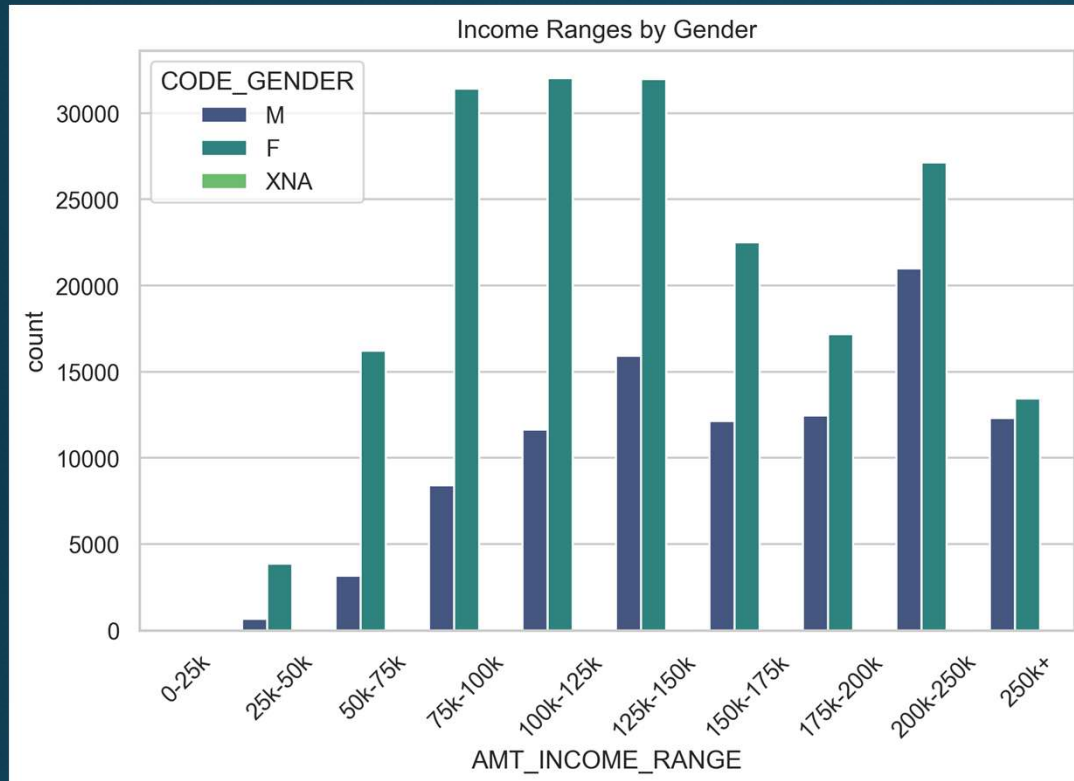
The class imbalance calls for resampling or some weighted modeling to better develop a predictive analysis.

credit distribution



- **Credit Amount Spread:** The histogram shows the distribution of credit amounts in the dataset, highlighting common loan sizes.
- **Peak Observation:** A significant concentration of credit values is observed within a specific range, indicating typical loan preferences.
- **Business Insight:** Understanding credit distribution helps in designing loan products tailored to the most frequent ranges.

income ranges gender

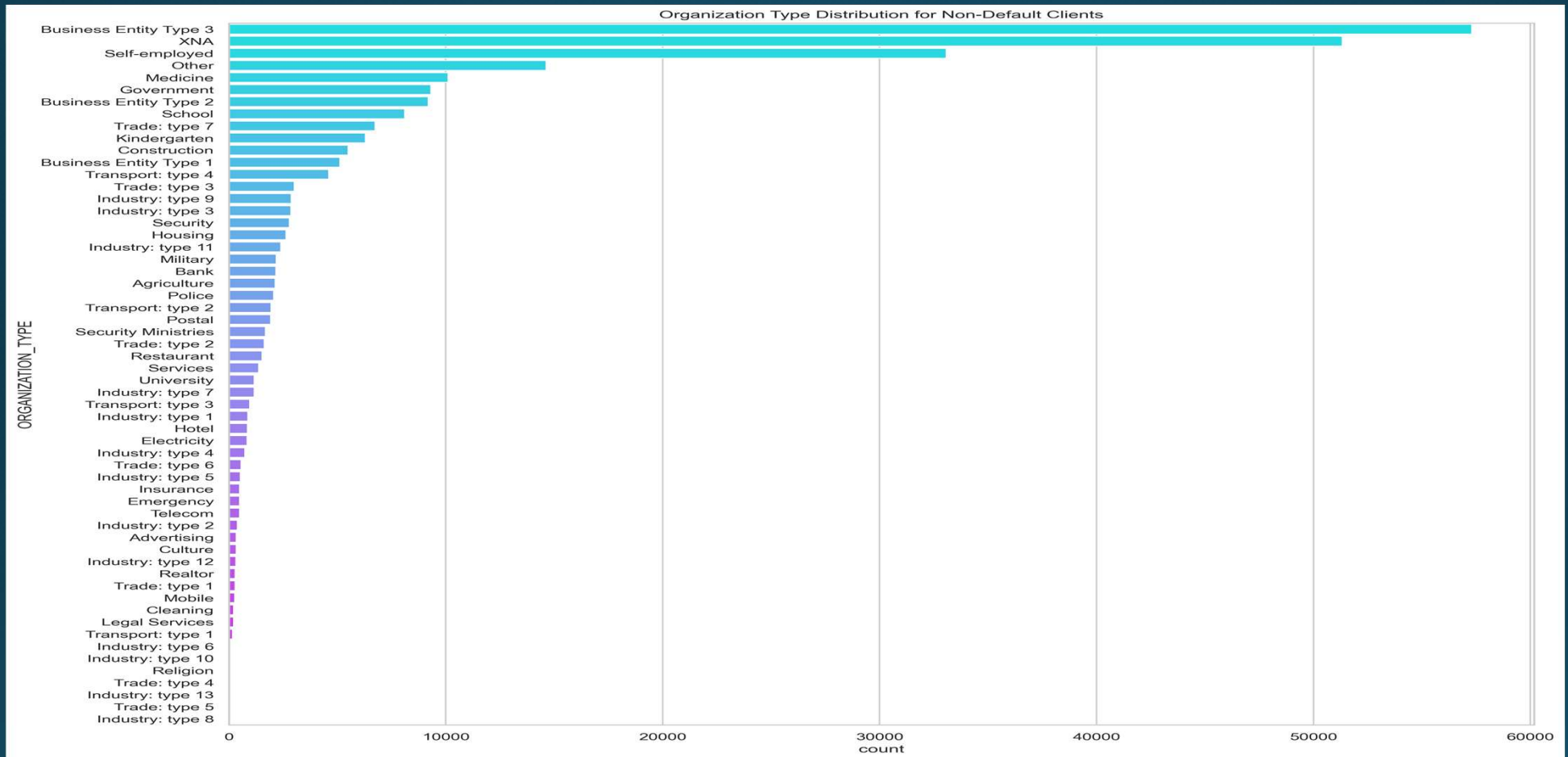


Gender-Based Income Distribution: The graph shows income groups categorized by gender, illustrating the distribution of income between genders.

Major Groups: There are some income groups, like middle-class groups, that have a greater percentage for both genders.

Key Takeaway: This analysis helps in understanding the trend of income between genders, which is very important for gender-specific financial planning.

organization types non default clients



An in-depth look into various types of organizations that the clients are affiliated with, along with their distribution between the defaulters and non-defaulters. This kind of association can be helpful for a prediction of client behavior and financial stability.

Conclusion

Key findings:

- Default clients show distinct correlation patterns compared to non-default clients.
- Income distribution reveals significant ranges, with gender-based differences.
- Credit and organization type also influence default likelihood.

THANK YOU